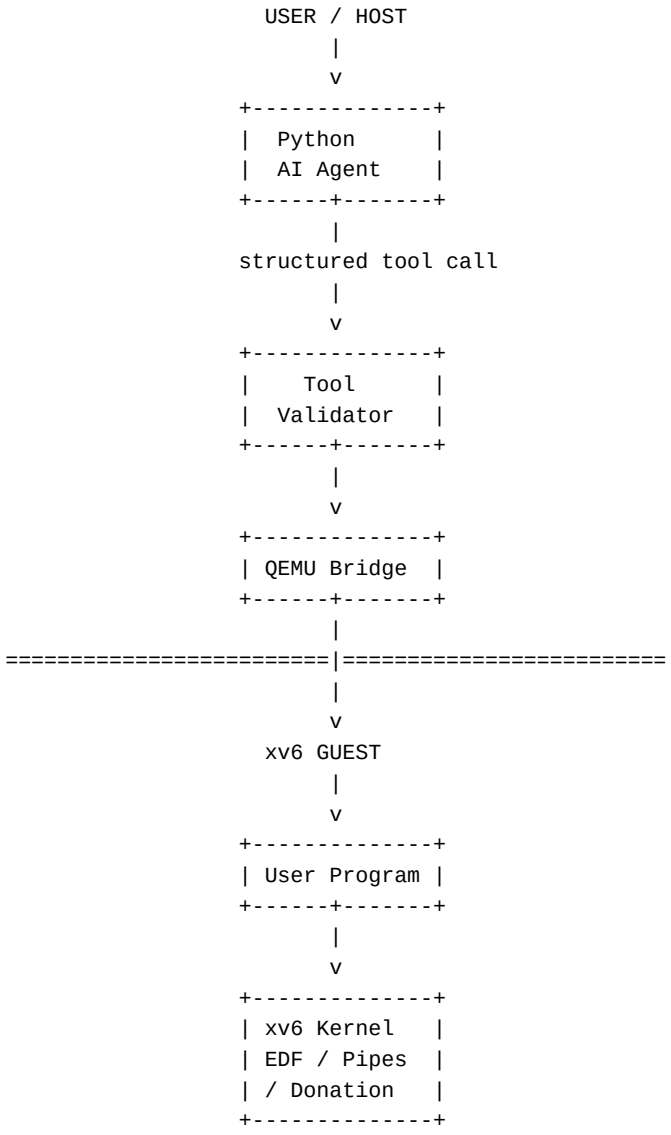
OS Major — Updated Mental Model

Group 4 • Deadline Scheduler with Priority Donation Through Pipes • Including the AI Agent + xv6 layer

1. The Big Picture

The project now has two connected layers. The OS layer is responsible for scheduling and kernel behavior. The AI-agent layer runs on the host and interacts with xv6 through a controlled bridge. The agent should not become the scheduler; xv6 remains responsible for enforcing OS policy.



2. Process States

State	Mental model
RUNNING	The process currently has the CPU and is executing.
RUNNABLE	The process is ready to run but is waiting for the CPU.
BLOCKED / SLEEPING	The process cannot currently proceed because it is waiting for an event, data, or resource.

3. The Most Important Distinction

```
RUNNABLE
= "I can run, but I am waiting for the CPU."

BLOCKED
= "I cannot run yet because I am waiting for something."

Only RUNNABLE processes are candidates for the scheduler's next choice.
```

4. Process State Flow

```
             gets CPU
RUNNABLE -----------------> RUNNING
   ^                          |
   |                          | needs something
   |                          v
   +--------- wake -------- BLOCKED
```

5. Pipes and Blocking

```
P1 -----------------> PIPE -----------------> P2
       writes data                       reads data

If the pipe has no data:
P2 cannot continue
       |
       v
P2 BLOCKS / SLEEPS
       |
       | data becomes available
       v
P2 wakes -> RUNNABLE -> eventually RUNNING
```

6. Pipeline

A pipeline is a sequence of processing stages. In this project, processes can form a dependency chain through pipes.

```
P1 --------> P2 --------> P3 --------> P4
       pipe          pipe          pipe

Data / work flows forward through the pipeline.
```

7. Deadline and EDF

A deadline represents when work is intended to finish. EDF means Earliest Deadline First. Among RUNNABLE processes, the scheduler prefers the process with the earliest effective deadline.

```
RUNNABLE processes
        |
        v
compare effective deadlines
        |
        v
earliest deadline wins
        |
        v
     RUNNING
```

8. Deadline / Priority Donation

Donation propagates urgency backward through a dependency chain. If a downstream process has an urgent deadline and is waiting on another process, that urgency can be donated to the process it needs.

```
DATA FLOW:
P1 ----------> P2 ----------> P3 ----------> P4

URGENCY / DONATION:
P1 <---------- P2 <---------- P3 <---------- P4
                                              |
                                        deadline = 100
```

9. Effective Deadline

A process may have its own deadline and may also receive a donated deadline. The scheduler uses the effective deadline rather than blindly using only the process's own deadline.

```
P1:
    own deadline     = 500
    donated deadline = 100

effective deadline = min(500, 100)
                   = 100
```

10. The AI Agent Mental Model

The agent is a program that repeatedly selects structured actions, invokes tools, observes results, and decides what to do next. An LLM may provide planning, but tool definitions, validation, execution, observations, and stop conditions are part of the agent system.

```
User goal
   |
   v
Planner / Agent
   |
   v
Structured action
   |
   v
Tool validator
   |
   v
Tool execution
   |
   v
Observation
   |
   +------> planner again
   |
   +------> finish
```

11. Trust Boundary

Model-generated text is treated as untrusted input. The agent should produce structured tool calls, the validator should enforce an allowlist and argument checks, and the kernel remains the final OS policy enforcement point.

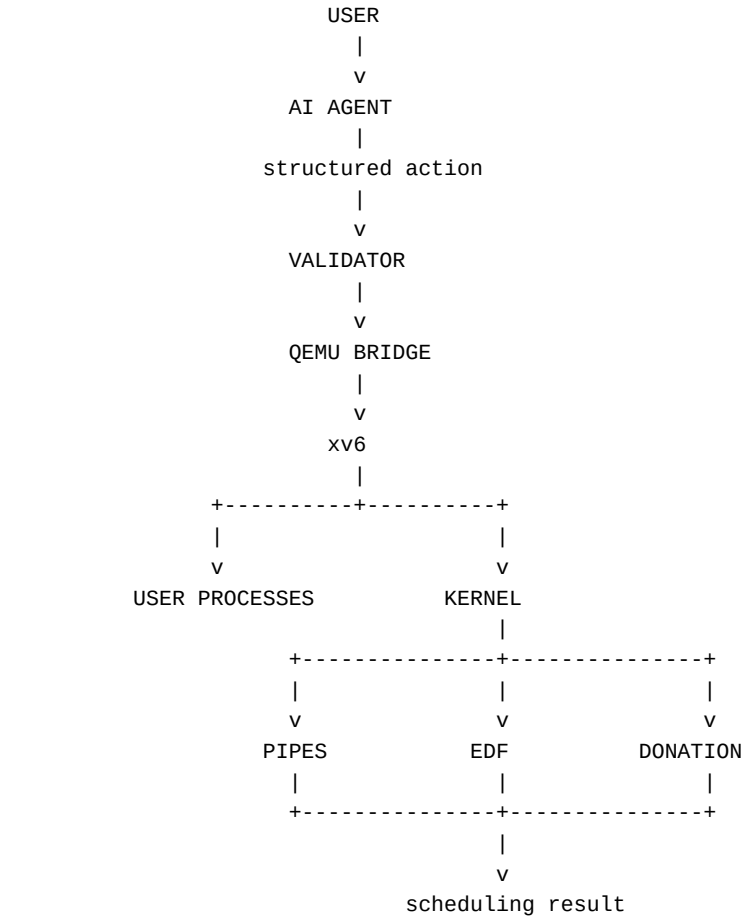
```
LLM / Agent
     |
     | proposes
     v
Tool Validator
     |
     | allows only valid structured actions
     v
QEMU Bridge
     |
     v
xv6
     |
     v
Kernel policy
     |
     +--> scheduler
     +--> pipes
     +--> deadlines
     +--> donation
```

12. What the AI Does NOT Do

```
BAD mental model:

LLM
 |
 +--> "Run P3 now"
 |
```

```
 +--> xv6 blindly follows it


CORRECT mental model:

Agent
 |
 +--> "This workflow has deadline 500"
 |
 +--> validated tool call
 |
 +--> xv6
        |
        +--> EDF scheduler decides
        +--> donation handles dependencies
        +--> kernel enforces policy
```

13. Complete Project Mental Model




```
KEY IDEA:
AI requests workflow/tool-level actions.
xv6 decides how processes are actually scheduled.
```

14. Development Flow

```
TRACK A — OS / KERNEL
Understand process states
        |
Understand xv6 scheduler
        |
Understand pipes
        |
Workflow + deadlines
```

```
        |
EDF
        |
Donation
        |
Chain propagation
        |
Tests + measurements


TRACK B — AI AGENT
Understand agent loop
        |
Deterministic Python agent
        |
Structured tools
        |
Tool validation
        |
QEMU bridge
        |
Connect to xv6
        |
Add LLM
        |
Agent observes xv6


FINAL INTEGRATION
AI Agent -> validated tools -> QEMU -> xv6
                                        |
                                        v
                                EDF + donation + pipes
```

15. Six Key Terms to Remember

Term	Meaning
Pipeline	Multiple processing stages connected sequentially.
Pipe	A communication channel used by processes to exchange data.
Blocking	A process cannot proceed and waits for an event/resource.
Deadline	The desired time by which work should finish.
EDF	Earliest Deadline First: prefer the earliest effective deadline among runnable processes.
Donation	Propagating urgency from a dependent process to a process it needs.

Final mental anchor: Data flows through the pipeline; urgency can flow backward through dependencies; the scheduler chooses among runnable processes; and the AI agent sits above xv6 as a controlled interface, not as a replacement for the kernel scheduler.